

Decision-Support Modeling Report

A leakage-safe machine-learning study of probability calibration, threshold selection, uncertainty, and model risk.

Final model	Sigmoid-calibrated Logistic Regression
Held-out test	54 observations (24 positive, 30 negative)
Recommended comparison rule	Default probability threshold: 0.50
Headline performance	ROC-AUC 0.908 Recall 0.917 F1 0.863

Central finding: the threshold that minimized illustrative cost during development did not generalize. On the held-out test set it produced two extra false positives without preventing an additional false negative.

Responsible-use notice

This report documents an educational machine-learning demonstration. It is not a medical device, diagnostic system, medical recommendation, or substitute for assessment by a qualified healthcare professional.

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1. Executive Summary

This project examined heart-disease screening as a cost-sensitive binary-classification problem. The aim was not to diagnose disease, but to demonstrate how an analyst can connect predictive performance to asymmetric error consequences, calibrated probabilities, threshold governance, and honest held-out validation.

The UCI Statlog Heart dataset was divided into a stratified development sample of 216 observations and a held-out test sample of 54 observations. Preprocessing was contained within scikit-learn pipelines. Logistic Regression, Random Forest, Gradient Boosting, and a Dummy baseline were compared using repeated stratified cross-validation.

Logistic Regression was selected because it offered the best overall balance of accuracy, recall, F1 score, calibration, and interpretability. Sigmoid calibration modestly improved the out-of-fold Brier score from 0.1238 to 0.1224.

An illustrative cost function assigned a relative cost of 1 to each false positive and 5 to each false negative. A threshold of 0.36 minimized estimated development cost, reducing cost from 134 to 79. However, the lower threshold failed on the held-out test: recall remained 0.917 while false positives rose from 5 to 7, increasing cost from 15 to 17.

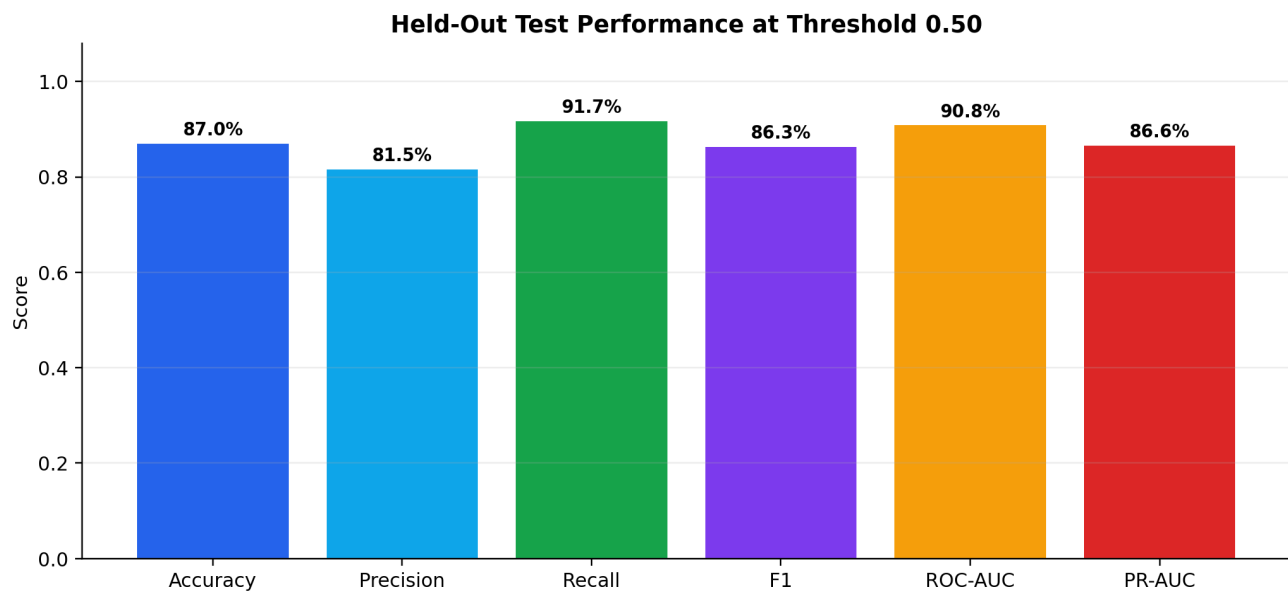


Figure 1. Final held-out performance at the default 0.50 threshold.

Recommendation

Do not adopt the 0.36 threshold. Retain 0.50 only as the better-performing prespecified comparison rule in this experiment, not as a deployment-ready clinical threshold. Any operational use would require larger, representative, external, temporal, and prospective validation.

2. Problem Definition and Data

Business question

Can a machine-learning model help prioritize individuals for further professional assessment while explicitly accounting for the higher illustrative cost of missing a positive case?

Decision context

In screening, error types are not interchangeable. False negatives may delay further assessment, while false positives may create unnecessary follow-up, cost, and anxiety. The project therefore reports discrimination, precision, recall, F1, calibration, error counts, and an explicit cost comparison.

Dataset

The analysis uses the UCI Statlog Heart dataset: 270 observations, 13 predictors, and a binary indicator of heart-disease presence. The features cover demographics, symptoms, physiological measures, exercise-test results, and diagnostic-test variables.

Partition	Rows	Disease present	Disease absent
Development	216	96	120
Held-out test	54	24	30
Total	270	120	150

Illustrative cost framework

Error type	Relative cost	Interpretation
False positive	1	Unnecessary additional assessment
False negative	5	Potentially missed positive case

The 5:1 ratio is an analytical assumption, not a clinically validated utility estimate. It is used to demonstrate cost-sensitive decision analysis and must be replaced by stakeholder-approved values in any real application.

Core limitations

- Small, historical dataset with only 54 held-out observations.
- No external, temporal, or prospective validation.
- Limited ability to assess subgroup fairness.
- Historical variables may not reflect current practice or target populations.
- Observational feature importance is neither causal nor diagnostic.

3. Modeling and Validation Design

Leakage controls

The held-out test set remained isolated until final evaluation. Numerical predictors used median imputation and standardization. Categorical predictors used most-frequent imputation and one-hot encoding. All transformations were contained inside model pipelines so validation folds did not learn preprocessing parameters from their own assessment data.

Repeated cross-validation

Models were evaluated with five-fold repeated stratified cross-validation using 5 repeats. This produced 25 validation estimates per model and reduced dependence on a single fold assignment.

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	Brier
Logistic Regression	.831	.818	.812	.808	.908	.909	.123
Random Forest	.827	.821	.796	.802	.914	.916	.124
Gradient Boosting	.810	.806	.771	.782	.888	.884	.135
Dummy Baseline	.556	.000	.000	.000	.500	.444	.247

Model selection rationale

Random Forest achieved slightly higher ROC-AUC and PR-AUC, but Logistic Regression delivered higher accuracy, recall, F1, and slightly better probability calibration. Its interpretability and simpler decision structure also supported selection for a screening-oriented prototype.

Calibration

Probability model	Out-of-fold Brier score
Uncalibrated Logistic Regression	0.1238
Sigmoid-calibrated Logistic Regression	0.1224

The improvement was modest, but calibration was retained because threshold decisions depend on the quality of estimated probabilities, not only ranking performance.

4. Threshold Analysis and Held-Out Test

Development threshold selection

Rule	Threshold	Precision	Recall	F1	FP	FN	Cost
Default	.50	.837	.750	.791	14	24	134
Minimum-cost	.36	.748	.896	.815	29	10	79

During development, lowering the threshold to 0.36 improved recall and reduced the illustrative cost by approximately 41%. The threshold was selected without using held-out test labels.

Final held-out results

Rule	Thr.	Acc.	Bal. acc.	Prec.	Recall	Spec.	F1	FP	FN	Cost
Default	.50	.870	.875	.815	.917	.833	.863	5	2	15
Cost-sensitive	.36	.833	.842	.759	.917	.767	.830	7	2	17

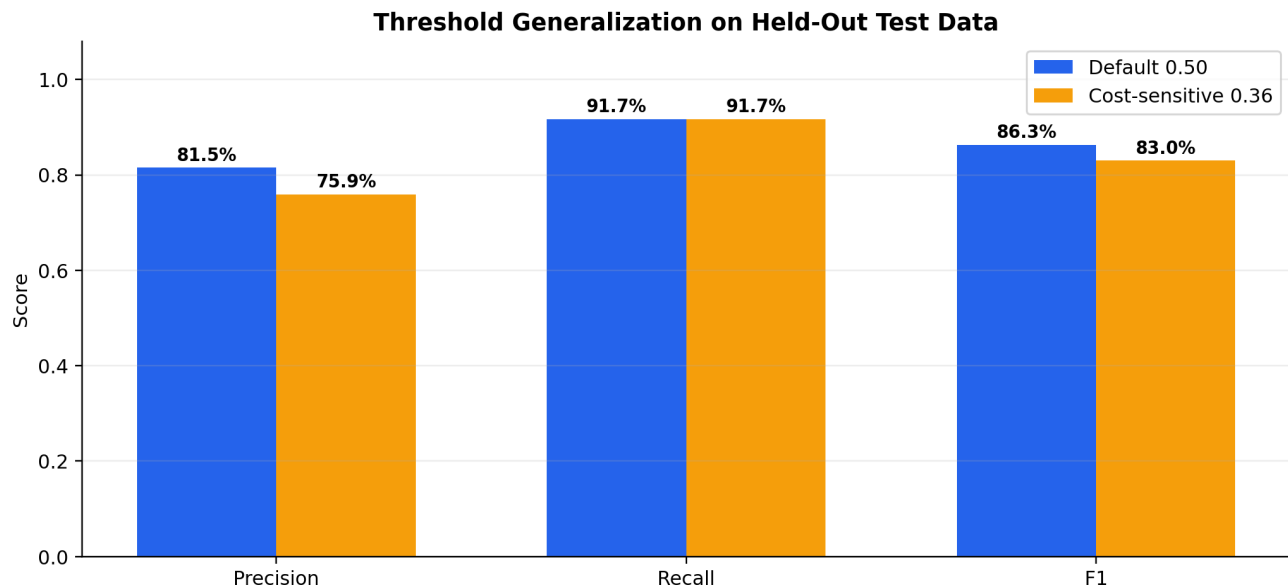


Figure 2. The lower threshold did not improve recall on the held-out sample.

Interpretation

Both thresholds detected 22 of the 24 positive cases. The lower threshold created two additional false positives but did not reduce the two false negatives. Held-out cost therefore increased by 13.3%. This is evidence that threshold optimization can overfit even when the model itself performs well.

5. Uncertainty and Interpretability

Bootstrap confidence intervals

Two thousand bootstrap samples were used to characterize uncertainty around default-threshold held-out results.

Metric	Estimate	95% CI lower	95% CI upper
Accuracy	.870	.778	.944
Precision	.815	.667	.957
Recall	.917	.792	1.000
F1	.863	.755	.949
ROC-AUC	.908	.815	.981
PR-AUC	.866	.717	.979
Brier score	.114	.069	.167
Relative cost	15.000	3.000	30.025

The intervals are wide because the test sample is small. Point estimates should not be presented as stable population performance.

Permutation importance

Feature	Mean importance	SD
Chest Pain Type	.0293	.0287
Major Vessels	.0216	.0334
ST Depression	.0186	.0190
Resting Blood Pressure	.0171	.0177
Serum Cholesterol	.0047	.0143
Thalassemia	.0045	.0285

These estimates show substantial variability. They describe sensitivity of this fitted model on this small test set; they are not causal effects or clinical rankings. Negative importance values observed for some variables indicate sampling instability or redundancy, not protective effects.

6. Conclusions and Recommendations

What worked

- Leakage-safe preprocessing and isolated held-out evaluation.
- Repeated model comparison across discrimination, classification, and calibration metrics.
- Modest calibration improvement with sigmoid calibration.
- Strong held-out discrimination: ROC-AUC 0.908 and PR-AUC 0.866.
- High held-out recall of 0.917 at the default threshold.
- Explicit documentation of uncertainty and model-risk findings.

What did not generalize

The development-optimized 0.36 threshold did not reduce held-out false negatives. It increased false positives and total illustrative cost. The correct analytical response is to reject that threshold, report the failed generalization, and avoid refitting a new threshold to the small test sample.

Recommended next steps

1. Validate on a larger, modern, representative dataset.
2. Pre-register the error-cost framework with qualified stakeholders.
3. Conduct external, temporal, and prospective validation.
4. Evaluate performance and calibration across demographic subgroups.
5. Define governance, human-review, appeal, and monitoring procedures.
6. Monitor calibration and data drift before and after any pilot.
7. Reassess clinical relevance of predictors and missing variables.

Final assessment

The project demonstrates competent end-to-end analytics and, more importantly, disciplined model governance. The final model is promising as an educational prototype, but the evidence is insufficient for clinical deployment. The most valuable result is not a single performance score: it is the demonstrated need to validate threshold choices on truly unseen data.

This report and its interactive demonstration are for education and portfolio presentation only.

References and Project Access

Dataset

UCI Machine Learning Repository. Statlog (Heart) dataset.

<https://archive.ics.uci.edu/dataset/145/statlog+heart>

Software documentation

Scikit-learn documentation: model evaluation, probability calibration, pipelines, and permutation importance.

<https://scikit-learn.org/stable/>

Project repository

[GitHub project folder](#)

Portfolio

gerald-hlabiso.github.io/business-analytics-portfolio

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Report scope

This report summarizes the validated results preserved in the project notebook. Numbers are rounded to three decimal places unless otherwise stated. Relative costs are illustrative analytical units.